

NETWORK THREATS AND PHYSICAL SECURITY RISK

1. Create a table listing 5 common network threats (such as phishing, DDoS, man-in-the-middle, malware, and password attacks), and for each, write one real-world example from an app or service you use (like WhatsApp, Instagram, or Paytm).

Network Threat	Real-World Example
Phishing	A fake Paytm login page steals your username and password.
DDoS Attack	Instagram becomes slow because attackers flood its servers with traffic.
Man-in-the-Middle (MITM)	An attacker intercepts data when using public Wi-Fi to access WhatsApp.
Malware	Downloading a fake app installs malware that steals personal data.
Password Attack	Someone repeatedly tries to guess your Instagram password.

2. Simulate a brute-force attack scenario by writing a simple Python script that tries to guess a 4-digit PIN code by checking all possible combinations against a preset value.

Constraint: The script should print the number of attempts it took to guess the correct PIN.

```
correct_pin = "1234"
```

```
attempts = 0
```

```

for i in range(10000):
    guess = f"{i:04d}"
    attempts += 1

    if guess == correct_pin:
        print("PIN Found:", guess)
        print("Attempts:", attempts)
        break

```

```

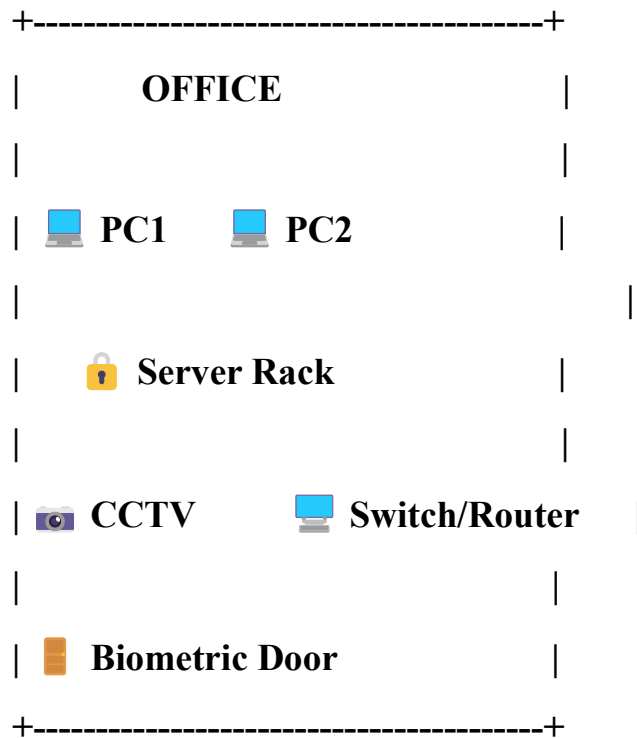
PIN Found: 1234
Attempts: 1235

```

3. List 4 physical security risks that could affect a server room hosting a popular app like Zomato or BookMyShow, and suggest one mitigation step for each risk.

Risk	Mitigation
Unauthorized entry	Install biometric or key-card access.
Fire	Install smoke detectors and fire extinguishers.
Theft of equipment	Install CCTV cameras and lock server racks.
Power failure	Use a UPS and backup generator.

4. Draw a simple floor plan (hand-drawn or digital) of a small office and mark at least 3 physical security measures you would implement to protect network equipment (such as CCTV, biometric access, or cable locks).



5. Use ChatGPT or a similar AI tool to generate a checklist of 7 actions a network admin should take to mitigate risks from both network threats and physical security breaches, then review and edit the checklist to make it specific for a college computer lab environment

- ✓ Use strong passwords for all computers and network devices.
- ✓ Install and regularly update antivirus software.
- ✓ Enable Windows Firewall on every computer.
- ✓ Allow only authorized students and staff to access the lab.
- ✓ Lock the server room with biometric or key-card access.
- ✓ Install CCTV cameras to monitor the computer lab.
- ✓ Regularly back up important files and keep operating systems and software updated.

Edited for a College Lab:

This checklist is specifically designed for a college computer lab where multiple students use shared computers. It helps protect both the network (firewalls, antivirus, updates) and the physical equipment (CCTV, locked server room, controlled access).